

Power BI Modeling

30-Minute Hands-On Exercise | Week 9 - Section 2B

Learning Objectives: Explain the structure of a Power BI semantic model | Distinguish data tables from lookup tables and apply correct naming conventions | Describe how relationships, cardinality, and granularity affect model accuracy | Configure column properties including data type, formatting, sort order, data category, default summarization, and navigational hierarchies

Instructions: Complete all four parts in 30 minutes. Refer to your Section 2B notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 2B.

1.	In Power BI, a data model is referred to as a <u>Semantic Model</u> .
2.	Tables that contain numeric measures and represent events (the center of a star schema) are called <u>Data or Facts table</u> .
3.	Tables that contain descriptive attributes used to group or filter facts (the outer tables of a star schema) are called <u>Lookup or Dimension tables</u> .
4.	Relationships between data tables and lookup tables should have <u>Many to One</u> cardinality -- many records on the data side, one record on the lookup side.
5.	The level of detail at which data is stored in a table -- for example, one row per transaction vs. one row per day -- is called <u>Granularity</u> .
6.	To make a Month Name column sort correctly (January before July), you configure a <u>Sort by Column</u> that points to a Month Number column.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term	Description
A. Semantic Model >> 5	1. Synonymous with a fact table; records describe events and contain measures
B. Cardinality >> 6	2. A metadata tag on a column that enables specialized visualization behavior (e.g., mapping)
C. Navigational Hierarchy >> 3	3. A drill-down order added to visualizations, such as Year > Quarter > Month > Day

D. Data Table	>> 1
E. Default Summarization	>> 4
F. Data Category	>> 2

4. The aggregation type Power BI applies to a column when it is dragged into a visual (e.g., Sum)
5. Power BI's term for a data model; built on the xVelocity engine
6. The explicitly defined numerical relationship between two sides of a model relationship (e.g., many-to-one)

Part 3: Column Properties (~8 minutes)

For each situation below, identify which column property you would configure. Choose from: Data Type | Numeric Formatting | Date/Time Formatting | Data Category | Sort By Column | Default Summarization | Navigational Hierarchy.

#	Situation	Column Property to Configure
1	A city name column is not rendering on a map visual even though the data looks correct.	Data Category
2	A 'Popularity Rating' column (values 0-5) is being summed by default, which produces meaningless totals.	Default summarization
3	A 'Month Name' column sorts alphabetically (April, August, December...) instead of chronologically (January, February, March...).	Sort by Column
4	A column imported from a CSV is stored as text but contains date values that need to support time-intelligence calculations.	Data Type
5	A revenue column needs to display values as \$1,200.00 with a dollar sign and two decimal places.	Numeric Formatting
6	You want report users to be able to click into a map visual and drill from Country down to Region, then City.	Navigational Hierarchy

Part 4: Scenario Analysis (~9 minutes)

Read each scenario and apply a concept from Section 2B. Write the relevant concept and a brief explanation in the right column.

Scenario	Your Answer + Brief Explanation
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1. A developer names their tables 'FactSales' and 'DimCustomer' in the Power BI semantic model. A colleague says this is a problem. What naming convention should be used instead, and why?	Use business-friendly names such as Sales and Customer instead. Power BI recommends avoiding technical prefixes like Fact and Dim because report builders and business users understand simple names more easily.
2. An analyst joins a daily budget summary table directly to a transaction-level sales table. Revenue totals suddenly look 30x higher than expected. What modeling concept explains this problem?	Mismatch in granularity. The budget table contains one level of detail while the sales table contains another. When the tables are joined incorrectly, records can be duplicated and totals become inflated.
3. A data model has a relationship drawn directly between two data (fact) tables -- an Orders table and a Returns table. What schema rule does this violate, and how should it be fixed?	The star schema rule. Fact tables should connect through shared lookup tables (such as Date, Customer, Product) rather than directly to each other.
4. A report displays a column called 'Customer Name Amount.' A stakeholder is confused by the label. Using Section 2B naming guidelines, identify at least two problems and suggest better names.	Problems: the name is unclear, it combines multiple concepts. Better names: Customer Name, Sales Amount. Each column should have a clear purpose.
5. A product manager wants to click on a bar chart and drill from Category down to Size and then Color. What Power BI feature would you configure to enable this, and what would the hierarchy look like?	Navigational Hierarchy. This will allow the user to drill deeper into the data.

Bonus / Reflection (if time allows)

Section 2B states: 'Most difficult data analysis problems are rooted in a flawed model.' In 3-5 sentences, explain what this means using a real or hypothetical example. What kinds of report errors or incorrect numbers could a poor model cause, and how does following star/snowflake schema rules help prevent them?